



Toward More Inclusive Music Experience: Understanding Deaf and Hard-of-hearing Individuals' Everyday Music Activities

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Abstract

Music can play an important role in the lives of some Deaf and Hard-of-Hearing (DHH) individuals, facilitating emotional expression, storytelling, and social interaction despite differences in hearing ability and identity. While prior human-computer interaction (HCI) research has introduced various functional advancements to enhance their music experiences, a deeper exploration of broader user experiences and inclusive design strategies remains necessary. To address the real-life challenges DHH individuals face in everyday music activities, we conducted focus group interviews in South Korea with 39 DHH individuals and 9 music experts. Our analysis identified six dimensions of everyday music activities organized by engagement type and social level, highlighting the distinct challenges and preferences DHH individuals encounter in musical contexts. Based on these insights, we propose design implications for fostering more inclusive music experiences, extending beyond individual engagement to include community and mixed-group interactions. This work provides a comprehensive framework to inform future HCI research and guide the development of inclusive technologies that better support DHH individuals' diverse musical experiences.

CCS Concepts

• **Human-centered computing** → Empirical studies in accessibility; • **Social and professional topics** → People with disabilities.

Keywords

Deaf and Hard-of-Hearing (DHH), inclusive music experience, inclusiveness, accessibility, disability, focus group interview, qualitative study

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1 Introduction

Across the diverse spectrum of identities and degrees of deafness [13, 56, 63], music holds significance for DHH individuals not only as a means of emotional expression and communication, but also as a medium of artistic engagement and cultural belonging. As such, music is an integral part of their everyday lives. DHH individuals actively construct their own sensory pathways for experiencing music by drawing on visual and tactile modalities as well as residual hearing [41]. Through vibrations, rhythms, and visual cues, they build unique musical experiences, engage in performances by singing or playing instruments, and even combine music with their distinct language (e.g. sign language) to establish new artistic domains [70]. They engage with music as part of their everyday experiences, as they sing along with their voices, move their bodies to the rhythm, and invest time and resources in enjoying musical performances [24, 25, 80]. For children in particular, musical activities play a crucial role in the development of language and cognitive abilities, highlighting the importance of including music education in the curriculum for DHH children [69, 73, 81].



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In the field of human-computer interaction (HCI), researchers have proposed various technologies to enhance the music experiences of DHH individuals by conveying musical information through alternative modalities. These approaches include sensory augmentation [5], multi-modal feedback [32, 72], and lyrics translation [2, 64, 67]. Among these, haptic feedback—delivering vibrations through the tactile modality—has garnered significant attention due to its natural resemblance to auditory sensations. Commercialized examples, such as musical haptic feedback embedded in car seats or smartphones, illustrate the increasing adoption and practical applications of this technology [43].

Although these technological advancements hold promise for improving the music experience, many investigations focus on controlled, laboratory settings and place substantial emphasis on music cognition. Consequently, these approaches have often overlooked the lived experiences of DHH individuals in everyday contexts. In response, recent work has begun to employ human-centered design methods, including focus group interviews [31, 80, 88] and participatory design [16, 85], to propose more inclusive, user-centered solutions. Despite these efforts, there is still a need for a comprehensive perspective that can demonstrate the areas of design that encompass challenges and opportunities of DHH individuals' everyday music activities, where cutting-edge HCI research can make significant contributions.

To address this gap, we conducted focus group interviews in South Korea with 9 DHH music experts and 39 DHH individuals to understand their lived experiences with daily music activities and to examine the challenges and opportunities in designing inclusive music practices. Through the interviews, we identified six dimensions that represent DHH individuals' everyday music activities, categorized by engagement type and social context. These findings highlight how DHH individuals engage with music, from sensory-based personal appreciation to community and mixed-group participation. Participants emphasized the need for alternative modalities like vibrations and visual cues, as well as linguistic supports such as lyric translations. They also noted the lack of DHH-centered music instruction and challenges in collaborating with hearing peers. Building on these findings, our discussion provides design implications for creating more inclusive and meaningful music experiences that address the diverse needs of DHH communities, framed through the lens of the HCI study.

The contribution of this study is threefold:

- **Diverse Participant Spectrum:** We conducted interviews with a wide range of deaf identities individuals and music experts who both perform and curate musical experiences, thus capturing a broad spectrum of perspectives.
- **Six Dimensions of Everyday Music Activity:** We identified six dimensions that characterize everyday DHH music activities, offering a structured understanding of key types of engagement and social contexts.
- **Design Implications:** We offer concrete design implications to guide the development of more inclusive music experiences across personal, community, and mixed-group contexts.

2 Related Works

2.1 Sensory Augmentation for Experiencing Music

Building on the concept of neuroplasticity—the brain's ability to reorganize itself by forming new connections in response to learning, experiences, or external stimuli [22]—HCI researchers have explored methods for conveying sensory information through alternative modalities, a practice known as sensory substitution. While early studies primarily focused on translating visual signals into tactile feedback [5], subsequent work expanded to include audio-to-tactile conversions [65, 66]. More recently, this research trajectory has been conceptualized under the broader term sensory augmentation, reflecting a transition from focusing solely on compensating for a missing sense to also enhancing or extending existing sensory capabilities.

In the context of music, most HCI researchers have focused on the ability of tactile sensory input to stimulate the auditory cortex [75] and the potential of vibrations to convey characteristics similar to auditory information [46]. Since the introduction of early haptic display concepts, such as the haptic chair by Nanayakkara et al. [65], and devices like the Emoti-Chair [45] and Skin Music [40], multi-actuator systems have been leveraged to deliver spatialized vibrotactile feedback. Similarly, wearable tactile displays including the Tactile Tone System [76, 77], the MuSS-Bits bracelet [71], Mood Glove [62], Haptic Sleeve [82], and Hedonic Haptic Player [7] highlighted the portability and user-centric design of music perception devices. Also, the LIVEJACKET [38] and CollarBeat [74] explored full-body and localized tactile feedback. Tactile composition tools, including the Vibrochord [9] and the Body:Suit:Score [86], further advanced the field by enabling real-time music interaction and immersive experiences. Collectively, these studies underline the diverse approaches to representing musical elements like rhythm, pitch, timbre, and melody through touch [32, 72].

Some researchers argue that tactile feedback is less intuitive than visual feedback approaches [41]. Farney et al. [35] introduced the Music Animation Machine, using dynamic visual elements to represent notes and rhythms for DHH users. Researchers such as Yang et al. [87] and Zhou et al. [91] demonstrated that computer-aided visualizations can enhance music performance and perception in DHH students. Kim et al. [51] further expanded on this idea by translating musical sounds from instruments like violins into visual forms such as flowerpots and plants for interactive education. Beyond education, Moodplay [3] offered a mood-based interface connecting emotions to musical attributes, while ViTune [27] synchronized on-screen visual effects with music, improving listening experiences. These studies demonstrate diverse visualization approaches that bridge the gap between auditory and non-auditory modalities for DHH individuals [48].

2.2 Lyrics Augmentation for Understanding Music

Lyrics are also an essential component of music as a verbal expression that delivers messages. For DHH individuals, however, understanding and engaging with lyrics can be challenging due to their reliance on auditory perception. To address this, researchers

have explored various systems and tools that translate lyrics into accessible formats.

One common approach for enhancing the accessibility of musical content involves visualizing lyrics through captioning. For example, Mori and Fels [64] investigated the use of animated lyrics to convey the emotional content of music for deaf and hard-of-hearing audiences. Their findings indicate that animated text captions can enhance comprehension and emotional engagement with musical content. Similarly, Neves [67] examined the challenges and strategies for effectively conveying the emotional and cultural dimensions of music through subtitles. Aleksandrowicz [2] configures the effectiveness of different subtitling methods for conveying emotions in film music. However, such approaches have been more extensively applied to video content than to music itself, with a focus on visually conveying non-verbal expressions, such as emotions and tones embedded in spoken dialogue [26, 39, 50].

Beyond text-based visualization, recent studies have emphasized the role of sign language, a unique and integral aspect of DHH culture. Song signing represents a distinctive form of musical expression that integrates sign language with musical elements such as lyrics, rhythm, and melody [23, 60]. Yoo et al. [88] investigated the values and perspectives of d/Deaf individuals on song signing, providing insights into creating culturally responsive performances that bridge the gap between d/Deaf and hearing audiences. Zhou et al. [90] further explored DHH individuals' use of sign language, visual and haptic cues, and preferences for non-lyrical, instrument-heavy music, while highlighting challenges in technology adoption and accessibility. Additionally, advancements in large language models (LLMs) and generative AI have introduced promising tools for facilitating the translation of lyrics into sign language, enabling DHH individuals to engage with music in more meaningful and confident ways [17, 89].

2.3 HCI Approaches for Inclusive Music

Based on the social model of disability—which locates disability in the environmental, attitudinal, and institutional barriers created by society rather than in an individual's impairment [1]—contemporary notions of inclusiveness have broadened through the adoption of accessibility and universal-design principles [30, 42, 47, 58]. Despite the increased attention to inclusiveness in HCI, Shinohara et al. [78] pointed out that many mainstream technologies remained inaccessible to people with disabilities. Likewise, Mack et al. [59] emphasized the need for HCI researchers to align their chosen research priorities with those of disabled communities. Designing for disability, therefore, involves not only addressing functional barriers but also understanding the social dimensions of how disabled individuals interact with technology, requiring deeper insight into their daily challenges and greater involvement of these users throughout the research process.

Inclusive music education aims to ensure that all individuals, regardless of ability, have equal opportunities to engage in musical learning [12, 57]. This aim converged with HCI accessibility research, resulting in a variety of studies that explore inclusive design in music education and participation. Earlier work typically focused on converting musical information—such as pitch and

timbre—into forms that DHH individuals could perceive, thereby enhancing their cognitive grasp of music [5, 32, 72]. More recent HCI research has adopted ethnographic approaches to achieve a deeper understanding of the social needs and cultural characteristics of DHH communities [15, 16, 88], yet much of this work concentrated on the distinctive musical culture within the DHH community (e.g., song signing). Consequently, relatively few studies have examined the barriers DHH individuals face in everyday musical activities.

Our study seeks to clarify the challenges and design opportunities associated with the broad spectrum of daily musical activities in which DHH individuals engage, viewed from an HCI perspective. We further propose strategies for adopting inclusive design approaches to address these challenges and discuss the design implications.

3 STUDY DESIGN

This study seeks to understand the experiences and challenges of DHH individuals in engaging with music in their daily lives, spanning from the personal layer to broader social contexts. To this end, we conducted two focus group interview studies aimed at examining diverse experiences and voices related to the everyday music activities of DHH individuals: **(1) focus group interviews with DHH music experts** and **(2) focus group interviews with DHH individuals**. We tailored the interview protocols for each group—experts and DHH individuals—and conducted the interviews separately to better elicit their unique perspectives. We also provided a group environment where participants could share their opinions with peers, in turn fostering their autonomy and broadening their perspectives. The study protocol was approved by the Institutional Review Board (IRB No.KNU-HR2023006).

3.1 Focus group interviews with DHH music experts

The purpose of the expert interviews was to identify the professional practices in planning and facilitating music activities for DHH communities and the challenges they faced in the process.

3.1.1 Participants. We sought to recruit DHH music experts who had experience in facilitating music activities, education, or performances for DHH individuals. The interview was advertised through flyers distributed at social welfare centers in local universities and online communities where Korean DHH individuals and their associates actively communicate. A total of nine DHH music experts (E1–E9) expressed their willingness to participate and gave their informed written consent. Table 1 summarizes the demographics of the expert participants, including their expertise, degree of deafness, and occupations. The participants included two special education teachers at a deaf school, two music teachers at deaf schools, two members of organizations for Deaf music activities, and three Deaf music performers. On average, the experts had 16 years of experience (ranged 4–38 years) and all possessed practical experience working closely or collaboratively with DHH individuals to foster accessible and enriching musical experiences for DHH communities.

ID	Age	Gender	Category	Deafness	Occupation
E1	60	F	Deaf School Teacher	-	Director, National Center for Deaf Education Support (38yr)
E2	38	M	Deaf School Teacher	-	Teacher, School for the DHH (28yr)
E3	61	M	Deaf School Music Teacher	-	Music Teacher, School for the DHH (8yr)
E4	43	M	Deaf School Music Teacher	-	Professor, Department of Music (12yr)
E5	49	M	Deaf Music Group	-	Director, Deaf Music Group (14yr)
E6	42	F	Deaf Music Group	-	Adjunct Professor (10yr)
E7	25	F	Performing Arts	Deaf/deaf	Artist (Song-signer) (5yr)
E8	29	F	Performing Arts	Deaf/deaf	Artist (Actor) (4yr)
E9	56	F	Performing Arts	-	Representative, Art Director (25yr)

Table 1: Demographics of expert interview participants

ID	Deaf Identity	Language	Average Age (SD)	Gender (F/M)
F1-P1~P9	Deaf/deaf	Sign Language	28.4 (8.7)	4/5
F2-P10~17	Deaf/deaf & Hard-of-hearing	Sign Language	35.0 (11.9)	5/3
F3-P18~P25	Deaf/deaf & Hard-of-hearing	Bilingual (Sign, Spoken)	15.9 (0.4)	3/5
F4-P26~P34	Hard-of-hearing	Bilingual (Sign, Spoken)	35.8 (8.9)	5/4

Table 2: Demographics of FGI participants

3.1.2 Procedure. All nine expert participants took part in an in-person interview as a single group. One researcher served as the interview moderator, while four researchers participated as observers, taking on roles as timekeepers and monitoring group dynamics and non-verbal cues—such as signs of agreement or hesitation—through detailed note-taking. These observational notes were later combined with the recorded audio to produce a complete transcript. A sign language interpreter was also present to facilitate smooth communication between Deaf and hearing participants and the moderator. The interpreter's translations were incorporated into the transcript to ensure accurate representation of all contributions. Observational notes were later combined with the recorded audio to produce a comprehensive transcript.

We began by asking participants about their professional experiences and expertise in facilitating DHH music activities, as well as the key factors they considered important during these processes. We then asked the specific musical concepts and elements they aimed to convey to DHH individuals and the strategies they employed to do so effectively. Lastly, we asked about their experiences with using assistive technologies in DHH-focused music education, therapy, or performances, including their motivations to adopt such technologies and the challenges they faced.

The session lasted about 1.5 hours with audio recording, and we provided 200,000 KRW (approximately 150 USD) as compensation for their participation.

3.2 Focus group interviews with DHH individuals

This focus group interview aims to explore how DHH individuals understand music, make personal meaning from it, and incorporate musical activities into their everyday lives, while uncovering perceived or latent inclusive issues in it.

3.2.1 Participants. DHH participants were recruited based on the following criteria: individuals who (1) are aged 18 or older with hearing impairments, or under 18 with parental consent and accompaniment, (2) enjoy music and have experience engaging with or learning music in their daily lives, and (3) are able to communicate via sign language or text. A total of 34 DHH participants were recruited through the social welfare center of a local university. The average age of the DHH participants was 29 years (range: 16–55; SD = 11.4), with an equal gender distribution of 17 for both females and males (see Table 2). Before the study, we collected data on each participant's degree of deafness, Deaf identity, frequency of music engagement and primary communication method through a survey. Based on the survey results, we divided participants into four groups according to their self-reported Deaf identity and preferred communication methods, as these factors influence approaches to musical activities and cultural perspectives of DHH individuals [13, 56, 63]. The groups were categorized as follows: (1) F1: d/Deaf, Sign Language, Adult (2) F2: d/Deaf and HoH, Sign Language, Adult

(3) F3: d/Deaf and HoH, Bilingual, Youth, and (4) F4: HoH, Bilingual, Adult

3.2.2 Procedure. DHH participants participated in semi-structured focus group interviews lasting between 80 and 110 minutes. In all four sessions, the same researcher acted as the moderator, supported by four researchers as observers. Each session included a sign language interpreter and a text prompt system to facilitate communication between the moderator and DHH participants. As in the expert interview, the resulting transcripts were compiled by combining the interpreter's translations and the stenographer's notes to ensure precision.

The interview questions covered topics such as motivations for engaging with music, preferred or challenging music activities, experiences of sharing or enjoying music with others, desired musical activities, and suggested improvements in education or assistive technologies to foster inclusive music experiences for DHH individuals. All sessions were audio recorded, and participants received a compensation of 100,000 KRW (approximately 80 USD).

3.3 Data Analysis

All audio recordings from the interviews were transcribed and digitized using ATLAS.ti [4]. The qualitative data were analyzed through a thematic coding approach [10]. The first author began the analysis with open coding, creating an initial codebook. The second and third authors then critically reviewed the codes developed by the first author alongside the associated raw data and provided written feedback. Subsequently, all three researchers jointly conducted a comprehensive review of the codebook, discussing issues raised in the written feedback. Through this iterative process, codes were added, merged, or removed until differing views among researchers were reconciled.

During this process, we observed that DHH participants' music experiences were influenced by both the social level of the activity and the type of engagement. Drawing on literature in music education and psychology, we employ terminology that distinguishes passive (receptive) activities—primarily listening and learning—from active (performative) activities involving singing and instrument playing [28, 37]. We also recognize music's sociological dimension, given its potential to be performed and shared among various groups; thus, the social level becomes a critical factor, capturing whether one, two, or multiple individuals are involved [28, 83].

This led us to perform axial coding, restructuring the codes based on three social contexts: **(1) personal**, **(2) within the DHH community**, and **(3) mixed groups involving hearing individuals**, and two engagement types: **(1) Passive** and **(2) Active**. Within these overarching social context themes, the codes were categorized with a primary focus on: (1) DHH individuals' perceptions and attitudes toward current musical activities, (2) challenges encountered in engaging with music, and (3) strategies employed by DHH individuals and their associates to address these challenges.

4 FINDINGS

Below, we present our findings on the everyday music experiences of DHH individuals. Specifically, we introduce six dimensions of DHH's everyday music activities (see Figure 1), organized by their

social context (Individual, Community within DHH, and Mixed-Group with hearing individuals) and engagement type (Passive and Active): (a) *Intrinsic Appreciation*, (b) *Self-satisfying Expression*, (c) *Shared Learning & Appreciation*, (d) *Community Music Expression*, (e) *Incidental Musical Experiences*, and (f) *Collaborative Mixed-group Performance*. Table 3 summarizes the topics mentioned by participants during the interviews. Some of these topics directly align with the proposed activity categories, while others reflect participants' emotional, cultural, or behavioral perspectives on music engagement.

In the following sections, we discuss each of the three social contexts in turn. For each social context, we begin by describing the current musical experiences and practices of DHH participants within the passive and active engagement type, and then elaborate on the inclusiveness-related challenges that DHH individuals frequently encounter. Throughout these sections, P# (e.g., P1, P2) denotes a DHH participant, and E# (e.g., E3, E4) denotes an expert participant.

4.1 Individual Music Activities

4.1.1 Passive: (a) *Intrinsic Appreciation*. This area highlights personal music appreciation activities. Most DHH participants enjoyed music in diverse ways, driven by their own motivations and modalities, including residual hearing and other sensory channels or assistive devices used to create sensory stimulation.

... When I listen to music, it makes me feel calm, and I think it helped me go to school better. The subway itself is such a closed space, right? So it feels really suffocating. But listening to music during those times helps me. (P18)

While many DHH participants used auditory devices such as hearing aids and cochlear implants, some noted that the sound delivered through these devices was somewhat unpleasant and failed to resonate as music in their perception. This led them to explore and develop their own preferred ways of experiencing music, such as through unaided hearing, vibrations, or skin contact, rather than relying on auditory devices. For instance, P31 reported pressing their ears to speakers at maximum volume, touching speakers to feel vibrations, or sitting on vibrating devices while listening to music.

When I use hearing aids, it just feels like I'm hearing machine noises... like the speakers are tearing. So I just turn the speakers up as loud as they'll go and listen that way. (P31)

Beyond sensory appreciation, there was an interest in understanding song lyrics and engaging with the message conveyed through the music. Regardless of whether participants primarily used sign or spoken language, they expressed a shared preference for the verbal expressions in songs. They highlighted that understanding and connecting with the meaning of lyrics conveyed through subtitles or sign language interpretation was an essential aspect of engaging with music.

These days, there's a group called 'Sign Dandelion' that performs sign language poetry and songs. I often feel really touched when I watch them. (P1)

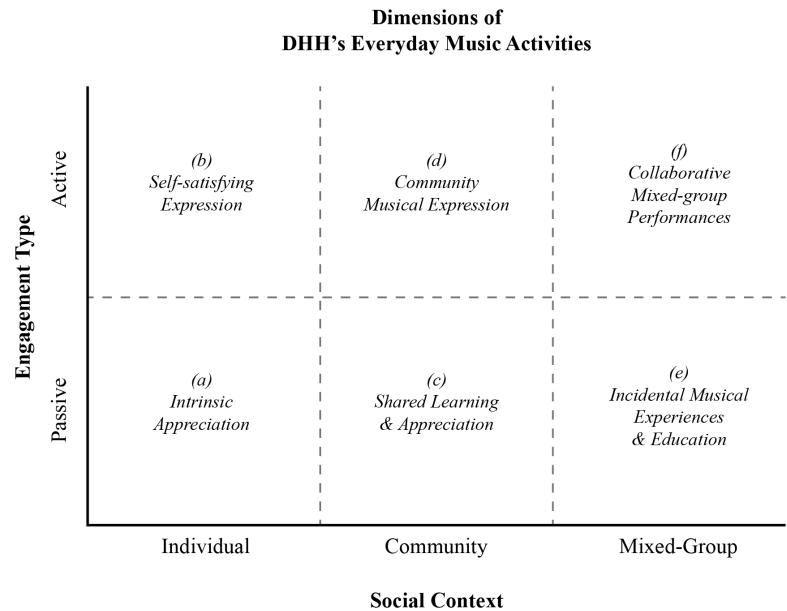


Figure 1: Dimensions of DHH's Everyday Music Activities

4.1.2 Active: (b) Self-satisfying Expression. This area encompasses activities where individuals actively engage in musical expression, focusing on sensory enjoyment rather than on flawless singing or playing. DHH participants often expressed music through dancing with excitement, moving rhythmically, or singing independently, with a particularly strong preference for singing and dancing. Some experts and DHH participants noted that DHH individuals sometimes worry that their ways of expressing music differ from those of hearing individuals, leading them to feel more comfortable enjoying these activities only in private spaces.

What I liked most was singing. When you sing, your vocal cords vibrate continuously, and I really liked that vibration. (P14)

Even if it's not perfect, they sing using a Deaf voice, like 'Ahh-'. They're fully enjoying themselves. But the thing is, they're worried about whether it sounds strange... They're enjoying it so much, but they're also conscious of how others might perceive them. (E2)

Participants also mentioned experiences of playing instruments like the piano, guitar, and bass. These activities were not about achieving musical proficiency but rather about expressing what they felt in the moment, fulfilling a more instinctive need.

4.1.3 Challenge: Connecting Verbal and Non-verbal Expressions. At the individual music activities, most DHH participants have already devised their own ways of enjoying music. Accordingly, rather than focusing solely on accurately perceiving sensory information, many participants indicated a strong need for understanding the lyrical content. This need for linguistic clarity extends beyond merely "reading" the song's message; participants also seek to perceive non-verbal musical elements such as rhythm and melody in tandem

with the lyrics. In other words, participants desire a harmonious delivery of music's non-verbal and verbal expressions.

When I listen to music, I always check the lyrics, because if I don't, I can't hear or know what the lyrics are saying. It'd be nice if I could understand the lyrics without looking and feel the rhythm at the same time. (P24)

When I went to the musical show, there was no way of understanding lyrics. So even though I spent money on it, I felt like I got nothing out of it. That's why I stopped going. But one day, there was a performance by foreign actors called Hekkan. I thought, 'Maybe they'll provide subtitles this time,' and decided it was worth buying a ticket to check it out. (...omitted...) I was able to understand what was going on, the context and situations, and even the tone of the voices and the conversations. Also, I could see their facial expressions. As a Deaf person, facial expressions are really important to me because they carry the emotions of the music (or songs), which makes it more fun. That time, I found the musical so interesting and enjoyable. I think that's what really matters. (P15)

4.2 Community Music Activities

4.2.1 Passive: (c) Shared Learning and Appreciation. This dimension describes music appreciation and education activities within DHH communities. While DHH participants were the primary agents of musical engagement in these environments, they often include hearing teachers or moderators facilitating the sessions. Such settings are commonly observed in special education schools or within DHH communities, where individuals learn and share their approaches of music appreciation. Activities include attending

Music Activity Category	Topics
(a) Intrinsic Appreciation	- Enjoying listening to music (P1, P3, P10, P12, P14, P18, P21, P23, P26, P28, P30) - Curiosity about appreciating music (P15, P29) - Discomfort caused by auditory devices (E7, P15, P25, P31) - Listening to music at maximum volume (P8, P11, P15, P27, P31) - Feeling music through vibrations (E7, P3, P5, P14, P15, P34) - Experiencing music through sign language (E2, P1, P15, P17, P24) - Need for subtitling lyrics (P5, P9, P11, P15, P23, P30, P31)
(b) Self-satisfying Expression	- Music is natural expression of emotion (E1, E2, E8) - Singing is enjoyable (E2, P5, P15, P23, P28, P30, P31) - Enjoying piano performances (P5, P30) - Enjoying drum performances (P15, P17, P30) - Enjoying guitar performances (P11, P24) - Enjoying violin and cello performances (P23, P34) - Enjoying flute and recorder performances (P9, P31) - Feeling stress relief and emotional stability by listening music (P11, P15, P17, P30, P34) - Feeling of vibrations on the fingertip is fun (P15, P24) - Enjoying creating sounds on my own (P9, P11, P15, P24)
(c) Shared Learning & Appreciation	- Music classes are challenging (E3, P9, P11, P15, P24) - Music is difficult because the teacher is not deaf (P23, P24, P31) - Learning and mimicking how other deaf people enjoy music (P3, P15, P31) - Learning rhythm is easy (E8, P5, P9, P15, P16, P23, P25, P26) - Enjoying music with deaf friends or family (P3, P9, P15, P19, P24, P27, P34) - Education feels like training (P9, P10, P15, P21, P26, P31)
(d) Community Musical Expression	- Enjoying going to karaoke with DHH friends (P7, P10, P15, P16, P17, P27, P34) - Motivated by other DHH individual's music activities (P4, P28) - Thinking karaoke is hearing culture (P9, P10, P15, P21, P34) - Music education encouraging expression through bodily movements (E3, E8, P3) - Expressing music through hand gestures and sign language (E2)
(e) Incidental Musical Experiences & Education	- Incidental situations where I have to listen to music (P5, P15, P16, P29, P30) - Difficulties from not knowing the music people talk about (P28, P29) - Inability to remember music (P2, P16, P19, P27) - Feeling isolated in hearing schools (P20, P27, P29) - Feeling excluded at music performances (P15, P17, P25, P29) - Enjoying musicals and concerts with subtitles (P17, P18, P25)
(f) Collaborative Mixed-group Performances	- Experiences in music education in hearing schools (P14, P18, P29) - Challenges faced during hearing culture (E5, P14, P25, P27) - Motivation for professional music experiences (P25, P29, P34) - Experiences with theatrical performances with hearing people (E7, P18, P25, P29, P34) - Communication difficulties during music performances (E7, P18, P25, P29)

Table 3: Interview topics answered by participants

performances together, exchanging thoughts on music in various settings, or engaging in music education programs, particularly in schools for the Deaf.

My entire family is Deaf. It's hard to distinguish sounds when listening to music through a hearing aid. But I had an aunt, who's also Deaf, who really loved music. I used to go to church with her and sing hymns, and it became something I enjoyed. I couldn't understand the rhythm taught at the Deaf school, but when my aunt

taught me the rhythm using gestures, it was a lot of fun.
(P3)

An interesting observation was the varied perception of similar musical concepts—some viewed them as “training”, while others experienced them as “education” or developed interest more naturally. Many participants felt as though they were being “trained”—intentionally or unintentionally—by the hearing music instructors at their schools to adopt music in the way hearing individuals do. An expert (E3)

also pointed out that not only do standardized music curricula tailored for DHH individuals remain insufficient, but the instructional system often reflect a hearing-centric perspective.

(After explaining their educational approach) We and the students always think that there's no other music room like ours anywhere in the country. It's the same for all general schools. They mostly focus on singing and playing instruments correctly, because the goal is preparing for university. On the other hand, special education schools should have different goals... (E3)

4.2.2 Active: (d) Community Musical Expression. This dimension refers to activities where groups of DHH individuals engage in musical expression together, such as singing, dancing, and other forms of performances. These activities were frequently observed in special education environments or within Deaf communities as part of their leisure activities. Participants shared experiences of ensemble performances and choir practices in church or school clubs, as well as memories of karaoke and arcade outings.

I think karaoke is great because the music is so loud. Even though I wasn't good at singing or keeping the pitch, I remember having so much fun trying. (P11)

Back then, we'd book one hour and get two or three hours of free time as a bonus. Our group was all DHH members—no hearing people. So, it didn't matter if I couldn't sing well, and it didn't matter if you couldn't sing well either. If it got too loud, I'd just take out my hearing aid. I think we once sang for 10 hours straight, all night long. (P34)

One notable observation was the recurring mention of karaoke experiences across all participant groups, which were often perceived as part of “hearing culture.” To overcome the barriers associated with engaging in hearing-centric spaces, participants expressed a preference for participating as a community rather than as individuals. They particularly enjoyed the unique setting of karaoke rooms, which provided loud music, real-time subtitles, and the opportunity to enjoy music collectively with other DHH individuals.

4.2.3 Challenge: Music Education for DHH, by the DHH. One of the most challenging aspects of music activities within the DHH community appears to be the limited opportunities for DHH individuals to impart musical practices rooted in their unique physical and cultural experiences to other DHH individuals. Our findings indicate that DHH individuals require instruction or guidance from educators who possess both cultural and technical understanding of their experiences. Ideally, this role could be filled by Deaf music teachers or other DHH individuals. However, some participants expressed concerns that hearing instructors often adopt inherently hearing-centered approaches to teaching music, which may not fully align with the needs and perspectives of DHH learners.

Wouldn't it be better if we stopped thinking that Deaf people can't be teachers? I think we need to get rid of this perception that it's impossible for Deaf individuals to teach. (P15)

Even if Deaf individuals want to join a singing class with hearing participants, there are no specialized instructors

who understand the needs of Deaf people. Because of that, it feels like we naturally fall behind when it comes to singing. (P14)

4.3 Mixed-Group Music Activities

4.3.1 Passive: (e) Incidental Musical Experiences and Education. This section addresses music appreciation and education activities in environments where hearing and DHH individuals coexist. A notable theme of this category is that DHH individuals engage with music as part of social and relational interactions with hearing individuals, often driven by external motivations and a sense of obligation. It includes situations where DHH individuals are required to participate in music education in hearing schools, everyday interactions involving music with hearing peers, and experiences attending diverse musical performances such as operas or musicals.

There was a recital where my younger sibling performed ballet, and it was really tough for me. Since my youngest sibling in the family is hearing, I sat in the back and endured until the very end... (P9)

My daughter is 36 months old now. I recently started listening and playing the ocarina for her because it's simple and easy for me to hear. Interestingly, when I sing songs from before I lost my hearing, my daughter enjoys them. But when I sing songs from after I lost my hearing, she tells me, 'Mom, please don't sing.' (P34)

Participants described incidental situations involving hearing family members who enjoyed music or performed and expected them to participate or engage. They also showed a willingness to learn about music to communicate better with their hearing family members.

I attended a hearing school. Since it was a hearing school, we learned both theory and practice in an integrated way, and the lessons were systematic, starting with theory. The theory was quite difficult for me, so I had trouble understanding them. Things like musical notes and rhythm were especially challenging for me. However, when it came to learning practical skills with the recorder, the teacher provided a lot of direct guidance, so I gradually started to understand more. I also tried hard to produce good sounds with the recorder and to keep up with the rhythm, which made it enjoyable. Eventually, I was able to perform together with all the other students. (P14)

Furthermore, many DHH participants who had attended hearing schools shared difficulties in music classes, such as understanding theoretical concepts, and expressed feelings of exclusion during discussions with hearing friends about movies or dramas that referenced symbolic music.

4.3.2 Active: (f) Collaborative Mixed-group Performances. This dimension addresses scenarios where hearing and DHH individuals perform music together. These include activities such as ensemble performances, choir singing, and theatrical acting, which demand a level of musical proficiency and collaboration beyond personal satisfaction.

Without a cochlear implant... I love music, but to truly experience it, I found that physically internalizing music first and then following rhythm and melody worked better. My hearing parents encouraged me to play piano, so I took lessons for four years. Instead of starting with sound, I learned rhythm and timing physically. For hearing people, vibrations and melodies might connect simultaneously, but for Deaf people, vibrations are more compelling. That's why it's important for Deaf individuals to first internalize vibrations physically before understanding rhythm and melody. (E8)

I'm currently an actor, and you know there are movies, plays, and musicals, right? Sometimes in acting, there are scenes where I have to sing. When I sing, I realized that it's not just about singing; I have to convey emotions too. And I discovered that there are all these variations in pitch. For the first time, I understood that music isn't just about singing—it's about expressing emotions, understanding pitch, and integrating rhythm and timing altogether. That's what makes it music. And that's what it means to be a true actor. (P25)

These types of musical activities require a higher level of musical perception, as participants must overcome unique challenges while striving for advanced performance standards. Many participants demonstrated their own methods and strategies for developing these skills, often exhibiting a strong intrinsic motivation and a deep preference for music. They showed little resistance toward structured music training, highlighting their dedication to mastering their craft despite the challenges.

4.3.3 Challenge: Music-based Social Interaction. The most significant challenge in these areas appears to stem from the communication barriers that arise when using music as a medium for interaction. Participants highlighted difficulties such as being unable to recognize the main theme music of films or dramas which are commonly discussed in everyday conversations, as well as challenges in communication while collaborating with other performers during musical activities.

I used to watch animated movies with my siblings and sing along. When they danced and sang enthusiastically, I felt excited too and tried to follow the songs. What I enjoyed the most was doing something musical together, like singing theme songs from cartoons such as 'Fly Superboard.' But personally, I always felt a bit left out. Hearing people can recognize these theme songs easily and say, 'Oh, this is a song I've heard before,' but I'd think, 'What is this?' Even if music played and everyone else sang along, I couldn't. (P28)

DHH participants engaged in professional music activities, despite their strong intrinsic motivation, frequently expressed frustration not with the musical performance itself, but with the communication difficulties that arose during collaboration. While they acknowledged the challenges of learning music within a hearing-centric framework, they noted that this could be overcome through repetitive practice. However, a more pressing concern was their

inability to fully perceive others' performances, which enables seamless collaboration.

The hardest part was not playing an instrument alone but playing in a group setting. It's not like I'm playing continuously from start to finish. Someone else plays, and when it's my turn, I have to join in, pause, and then play again. But I often don't know exactly when my turn is supposed to start. (P25)

When I'm on stage as an actor... For instance, knowing when to start singing, the timing for cues, or when to make an entrance—those things could probably be managed with a wrist device. But honestly, when you're on stage, the most important thing is the connection and rhythm shared between actors, not relying on devices. Reacting to devices can disrupt the flow, which is crucial for performers. So, as a Deaf actor, I feel like these tools don't really fit. (E7)

5 DISCUSSION

5.1 Bridging Barriers and HCI Approaches for Inclusive Music Experiences

Inclusiveness in music activities extends beyond artistic expression, striving to ensure that the cultural and social dimensions of music remain accessible to everyone [57]. Achieving this objective requires tackling the barriers that inhibit enjoyment and immersion for diverse populations [11, 20]. As Mack et al. [59] emphasized, HCI research focused on inclusiveness must attend to the real-world barriers encountered by individuals with disabilities.

Motivated by this perspective, our study identifies a variety of everyday obstacles that DHH individuals face when engaging with music, highlighting the need for more holistic HCI approaches that account for both functional and sociocultural factors. These include communication gaps within hearing-centric learning environments, insufficient support for community-based music activities, and constraints on collaborative performances in mixed groups.

These challenges and the corresponding dimensions of everyday musical activities reveal new opportunities for advancing inclusive music practices through HCI. Rather than simply adapting cutting-edge technologies, we advocate for a paradigm shift that centers on the lived experiences of DHH users. As Benett et al. insisted [6], participatory methods that actively involve disabilities, as demonstrated in work by Choi et al. [16] and Yoo et al. [88], appear especially well-suited for generating practical and context-specific inclusive technologies. Building on our findings to delineate clear domains, and then co-designing with DHH users according to their unique identities and experiences, can lead to more impactful solutions. By doing so, HCI can foster more meaningful musical engagement, moving beyond conventional notions of accessibility and allowing DHH individuals to fully explore, create, and participate in a wide range of music activities.

5.2 Design Implication

5.2.1 Individual context: Inclusive Music Augmentation. Individual-level music experiences and performances have been one of the

most frequently addressed contexts in the trajectory of HCI research. As highlighted in the literature review, many studies have targeted these areas by augmenting either the non-verbal sensory aspects or the verbal components of music.

Recent studies have increasingly focused on verbal expressions in music, such as song signing and lyrics visualization, which are identified as more challenging for DHH individuals than the sensory aspects. While there is significant demand for synchronized experiences that allow users to simultaneously feel the sensory and lyrical messages of music, only a few studies have suggested beyond subtitling lyrics [26]. However, approaches like font design for non-verbal expression in subtitles, as explored by Kim et al. [50], or haptic captioning to reinforce the connection between lyrics and music, as proposed by Wang et al. [84], demonstrated a potential of enhancing engagement with the verbal components of music. Furthermore, with advancements in natural language processing, higher-order translation technologies, such as generating sign language from song lyrics, have become feasible [18], marking this as an exciting area for future research.

In contrast, while sensory augmentation research has been conducted, haptic feedback remains one of the few technologies that has been commercialized [36, 43]. Oshiro and Cartwright [68] pointed out that DHH music creators often prefer the same visual interfaces as hearing users and do not seek additional assistive technologies. Similarly, interviews with DHH users reveal resistance to visibly notable assistive devices [79]. This suggests that future approaches should focus on lightweight and ambient technologies tailored to individual contexts.

Inclusive music augmentation is not limited to DHH individuals but offers a broader approach to making music's elements more intuitive and multi-sensory, extending to hearing audiences who also appreciate immersive listening experiences akin to live performances where visual effects and stage production play a critical role. However, as Choi et al. [15] showed, providing excessive sensory information can lead to confusion, emphasizing the need for customizable and balanced solutions tailored to individual preferences. Instead of solely aiming to improve cognitive recognition of musical information, technological approaches should help individuals explore and experience their preferred sensory and emotional aspects of music. Such experiential enhancements can foster motivation to expand into other areas of musical activity, making this approach particularly significant.

5.2.2 Community context: Inclusive Music Education. Inclusive music education has been extensively discussed in pedagogy [12, 57] and music science [69, 81]. However, our interviews revealed a significant boundary stemming from the dynamic of hearing instructors teaching non-hearing learners. This boundary underscores a broader cultural and communicative gap, as further illustrated in Yoo et al.'s work [88], where DHH participants expressed discomfort with music videos produced by non-Deaf individuals who used sign language to represent songs.

From an HCI perspective, one way to address this gap is by examining how technology can expand interaction in music education contexts, particularly when hearing instructors play a minimal role. For example, Muss-Bits [71] provides small audio receivers and modules that vibrate and emit light, enabling deaf students to

explore music collaboratively with minimal outside intervention. Also, researchers can focus on the DHH instructors to enhance their strategies [19]. Similarly, rather than adapting classical instruments for DHH use, designing custom digital instruments can foster intrinsic motivation and a stronger sense of ownership in musical engagement [33, 34, 61].

In summary, shifting away from hearing-centered methods that primarily aim to help DHH learners comprehend traditional musical concepts [21, 49, 53], and instead focusing on technologies aligned with DHH abilities and preferences, can create a more inclusive music-learning environment. Devices that support DHH-led musical interaction or digital instruments tailored to DHH needs can lower barriers to diverse musical activities, encouraging richer cultural and community participation.

5.2.3 Mixed-Group context: Inclusive Music Communication. Music activities at the mixed-group level are among the most common contexts DHH individuals encounter in daily life and are often essential for pursuing more advanced musical performance. These include acquiring musical knowledge to communicate with hearing friends or family members, as well as collaborating in ensembles with hearing peers.

One example addressing performer-to-performer communication is the Haptic Baton developed by Human Instruments [44]. This device enhances communication between conductors and DHH performers by capturing the conductor's hand movements and transmitting them as vibrations to performers, enabling them to perceive the conductor's timing and expression through tactile feedback. Such approaches demonstrate how technology can support communication during musical performances, highlighting a potential for further exploration in mixed-group music activities.

Furthermore, there is a growing need for research focusing on communication between hearing and non-hearing participants regarding musical language. Choi et al. [16] explored how collaborative creative processes involving DHH and hearing individuals can be supported through a multi-modal synthesizer. This system translated musical elements into visual and tactile feedback, enabling a shared understanding that reduced the reliance on interpreters and supported more seamless collaboration. Such systems can foster communication and enhance the participation of DHH individuals in hearing-centered music environments.

Despite the prevalence of such situations, existing research has primarily focused on how to convey a common language of music to DHH individuals, while paying limited attention to supporting communication around music within these contexts. Shifting the perspective from music perception to the communication dynamics between participants could significantly contribute to facilitating collaborative mixed-group activities. To achieve a more inclusive music experience, future HCI efforts should prioritize the development of technologies that facilitate communication across diverse user groups. These efforts could lower barriers for not only DHH individuals, but also other types of disability, and expand the possibilities for collaborative mixed-group musical activities.

5.3 Conducting Interviews with Deaf/deaf and Hard-of-hearing Participants

While a variety of interview methodologies have been proposed, most are predicated on spoken-language interaction. In our study, the main challenge was to conduct FGIs with multiple DHH participants in a way that supported natural turn-taking and ensured that every voice was heard.

Because participants relied on different communication languages—including sign language, spoken language, and bilingual—we employed both a professional sign-language interpreter and a stenographer to maximize fluent communication. However, this approach resulted in a larger number of people in the room and risked creating a formal and constrained atmosphere. To mitigate this effect, the moderator incorporated extensive ice breaking talks, grouped participants who shared similar Deaf identities, placed interpreters behind the moderator, and displayed real-time text prompts on a screen visible to all participants. Together, these measures minimized visual distraction and helped maintain participants' attention on the discussion.

Although prior work has begun to address these challenges [29], additional methodological research is needed to develop UX analysis techniques based on a deeper understanding of the communication practices of DHH. Recent group autoethnography in mixed hearing teams has shown that design tactics such as shared visual attention cues, pause and process signals, and multimodal backchannels can smooth real-time communication between hearing and DHH collaborators [14]. Based on these attempts, future HCI work can go beyond tool design to develop analysis protocols, thus broadening our methodological repertoire for inclusive research.

6 LIMITATION & FUTUREWORK

This study presents several potential limitations. First, although we recruited a relatively large and diverse group of DHH participants—thereby capturing a range of lived experiences—the FGI method has inherent constraints. Future work could complement FGIs with ethnographic approaches such as diary studies [8] and longitudinal observation [52], particularly for participants who exhibit strong motivation toward music activities. Such in-depth methods would allow researchers to observe participants' everyday musical experiences in context, potentially revealing more detailed insights into the distinctive features of each dimensions and the specific challenges DHH individuals encounter.

Second, while our analysis centered on DHH participants' diverse everyday music activities, we observed that some individuals also engaged in semi-professional or professional-level music pursuits. Although this paper primarily explores barriers and potential HCI approaches in everyday music activities, future research could investigate how professional music-making—such as songwriting, composing, or formal performance—might introduce novel opportunities and challenges for DHH communities. Studies focusing on Deaf musicians in more advanced creative processes could further enrich our understanding of inclusive design for both amateur and expert domains [19].

Lastly, Deaf culture is regarded as a distinct cultural community rather than simply another disability category, and variations may

arise across different national or regional contexts [54, 55]. Our study was conducted solely with participants residing in South Korea, and our findings should be interpreted with this context in mind. Researchers consider cultural, linguistic, and educational differences when applying our design spaces and implications to other settings.

7 CONCLUSION

This study examined the everyday musical experiences of DHH individuals by identifying six dimensions linked to the social context and engagement types of music activities. Our findings highlight the importance of social context, suggesting that community and mixed-group contexts deserve as much consideration as individual pursuits. Drawing on these insights from an HCI perspective, we propose design implications that include augmenting sensory and verbal elements, as well as exploring community-oriented and collaborative approaches. We encourage future research to refine and expand upon these possibilities—especially through participatory methods that center the lived experiences of DHH communities. In doing so, we hope to support more inclusive musical practices that allow DHH individuals to engage with, create, and share music in ways that reflect their diverse needs and aspirations.

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